

MIKHAEL KATON SURYO GUMILANG

NETWORK OPERATIONS ENGINEER

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SUMMARY

Telecommunications Engineering student with hands-on internship experience in FTTH network monitoring and troubleshooting using NetNumen. Experienced in handling backbone, OLT, and FAT network issues while developing strong problem-solving, teamwork, and communication skills. Also skilled in IoT-based monitoring system development and technical project coordination.

EDUCATION

Bachelor of Design in Process Engineering **2022 - Present**

Major: Department of Electrical Engineering; Telecommunication Engineering Study Program,
GPA 3.48/4.00

- Relevant coursework : Computer Networks, Telecommunication Systems, Data Communication, Antenna and Propagation, Transmission Line, Fiber Optic Communication System

PROFESSIONAL EXPERIENCE

FTTH Internship Program – PT Indosat

Semarang, Indonesia

FTTH Division

April 2025 – August 2025

- Performed FTTH network monitoring and troubleshooting using NetNumen.
- Handled troubleshooting related to backbone network disruptions, OLT, and FAT issues.
- Coordinated with technical teams to restore network stability efficiently.
- Assisted field operations and supported network maintenance activities.

PROJECTS

IoT-Based Aquascape Water Quality Monitoring and Control System

November 2025 - Present

Team Leader & Mechanical System Designer

- Integrated ESP32, sensors, relay modules, and monitoring interfaces for remote access through website and Android applications.
- Designed and managed the mechanical system while leading a team of four members.

Microstrip Antenna Implementation Projects

2023

Individual Project

- Designed and implemented microstrip antennas at 5.8 GHz and 1090 MHz frequencies using FCC and Alumina materials.
- Utilized CST Studio for antenna simulation, optimization, and performance analysis.
- Conducted manual fabrication, implementation, and testing to evaluate antenna characteristics and overall performance.

ADDITIONAL INFORMATION

- **Languages:** Native Indonesia, English Profesional certified by TOEIC.
- **Certifications:** CCNA: Switching, Routing, and Wireless Essentials, Introduction to Internet of Things (IoT)
- **Skills:** Network Monitoring, Network Troubleshooting, FTTH Network, Backbone Monitoring, NetNumen, Cisco Packet Tracer, Basic Understanding of OSPF, BGP, and MPLS, Problem Solving, Teamwork, Communication, Fast Learning.